

[L1] Chapters 1 & 2 - Prehistory

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- Before the 19th century, machines were used for their novelty / entertainment value
 - Some machines were just trickery
- In the early 19th century, the steam engine was invented
 - This proved that machines could serve a sensible purpose

Charles Babbage (1791-1871)

"On the economy of machines and manufactures"

- Mathematician who was interested in the economy of business and how to make business more efficient
- He was especially interested in using machines to achieve this
- Invented the Difference Engine and the Analytical Engine
- He wrote "On the economy of machines and manufactures"
 - A book about businesses he had visited and how to make them more efficient by removing bottlenecks (often by using machines)
- He came up with a non-mechanical solution too: **The clearing house**
 - A place where the bankers of London would meet and exchange checks
 - Babbage came up with the idea of replacing the massive clearing house on Lombard Street by having each bank open an account with the Bank of England
 - This transformed the problem into an accounting task
 - This is considered one of his greatest achievements
 - It took banks decades to adopt this => Even good ideas sometimes take time to implement

Frederik Taylor (1856-1915)

"Shop management" (1903)

"The principles of Scientific management" (1911)

- Designed principles that allowed manufacturers to identify and fix bottlenecks themselves
- Proposed more advanced labour division than Babbage had imagined
- Inspired Henry Ford, who invented the conveyor belt production line for cars

Henri Fayol (1841-1925)

- After learning of Frederik Taylor's work, he introduced it into French mines
- Looking deeper into the matter, he realised that French bureaucracy could benefit from the same process
- His efficiency optimisations are focused on the **administration** part of the business, while his predecessors focused more on the **production** side

Trust in numbers & machines

- These people were part of a larger movement, which was beginning to put more trust in machines and numbers in the early 20th century
- Proof: the US Census
 - By having data about the population, governments could predict tax income, how many people could be drafted in case of war, etc
 - 1890: Hollerith came up with machines to help with the counting and creating tables
 - Hollerith's machines not only did well, they did better than humans sometimes
 - This resulted in people trusting machines more and more
- The European continent was not as keen on machines as the US and the UK
 - When the **cash register** was introduced, it was embraced easily in the US, but Europe viewed it with

- some amount of skepticism
 - This was due to a more "leftish" political environment, but also due to the fact that companies had been around for longer and were used to doing things a certain way
 - Introducing new procedures was not something they were thinking about => tradition can hold progress back
- Industrialists and governments believed that machines could make life easier
- Automation was most desired in administration, process control and science
 - US focused more on the administration side
 - Europe focused more on process control and science

Administration

- The office in the late 19th century meant a lot of clerks writing things and copying things by hand
- In the early 20th century, this was replaced by typewriters
 - They could use carbon paper to have several copies typed at once
- Punched card machines were introduced for sorting, tabulating, paychecks
- Calculators, typewriters, sorting machines, counting machines, printers etc... were being introduced
- Using these machines, business administrators and owners could use Henri Fayol's and Frederik Taylor's principles in order to fix their businesses
- Not all businesses managed to successfully implement machines
- Not all of this was high-tech. High efficiency could be reached with filing cabinets too

Process Control

- There were processes that had to preferably be controlled from a distance (steel production, oil extraction and refining)
- This is a different kind of automation. Machines were using valves, sensors to take decisions themselves
 - Machines were being supervised, not operated

Science & Engineering

- Computers (PEOPLE!) were active in weather predictions, hydromechanical calculations, aeronautics, econometry, telephone cables, etc etc
- Douglas Hartree (1897-1958): "*Numerical Analysis*", Differential Analyser
 - These were analogue machines that were able to perform one type of calculation
- Most calculations were done on worksheets

The three traditions

- All three used outlines
- The administration business had the most money, so it is not surprising that that is where the first computer came from